

AhamX Bodhi

Where I Becomes Infinite

Text-to-Speech Systems

Module 4.2: Audio and Speech Models

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Session Overview

What You Will Learn

- TTS system components: acoustic + vocoder
- Architecture evolution: Tacotron to VITS
- Zero-shot voice cloning: VALL-E, Voicebox, XTTS
- Neural vocoders: HiFi-GAN, BigVGAN
- Engineering TTS for production applications

Concept at a Glance

- **Duration:** 9 minutes
- **Bloom's Level:** B2 – Understand
- **Difficulty:** L2 – Intermediate
- **Module:** 4.2 Audio and Speech Models
- **Key metric:** MOS (Mean Opinion Score)





TTS Pipeline: Text to Waveform

Classic Two-Stage TTS Architecture

- **Stage 1 – Acoustic model:** Text → acoustic features (mel-spectrogram). Handles grapheme-to-phoneme conversion, prosody, duration, and pitch prediction.
- **Stage 2 – Vocoder:** Mel-spectrogram → raw waveform. Inverts the mel-spectrogram back to audio, adding fine spectral detail that the compressed mel representation lost.
- **Text frontend:** Text normalization ("15" → "fifteen dollars") + grapheme-to-phoneme (G2P) conversion + prosody annotation – often underestimated in importance
- **Evaluation:** Mean Opinion Score (MOS) on 1–5 scale. Human speech: 4.5+. Modern neural TTS: 4.3–4.5. Older concatenative: 3.5–4.0.
- **Target:** Human-indistinguishable synthesis – zero-shot voice cloning models achieve MOS >4.0 in 2024.

Modern end-to-end TTS models (VITS, NaturalSpeech 3) skip the explicit mel-spectrogram stage and generate waveforms directly from text – collapsing two stages into one.



TTS Architecture Evolution

From Concatenative to Neural End-to-End

- **Concatenative TTS (pre-2016):** Stitch together pre-recorded speech segments. Robotic-sounding at boundaries, limited expressiveness. Festival, MaryTTS.
- **Tacotron (Google, 2017):** Seq2seq attention model: text → mel-spectrogram. First neural TTS to approach natural-sounding speech. Tacotron 2 + WaveNet: MOS 4.53 (near-human).
- **FastSpeech 2 (Microsoft, 2020):** Non-autoregressive Transformer – predicts duration, pitch, energy explicitly. 38× faster than Tacotron 2. Now the standard for production.
- **VITS (Jeong et al., 2021):** Variational Inference with adversarial learning + normalizing flows. End-to-end: text → waveform. Single model, no vocoder. MOS 4.43.
- **YourTTS / XTTS (2022–2023):** Zero-shot multilingual voice cloning based on VITS. 13+ languages, 3-second speaker reference.

VITS and its derivatives (VITS2, XTTS-v2) are the current open-source state of the art for production TTS – end-to-end, fast, and capable of voice cloning.



Neural Vocoders: Mel-Spectrogram to Waveform

From WaveNet to BigVGAN (2016–2024)

- **WaveNet (DeepMind, 2016)**: Autoregressive dilated causal CNN generating audio sample-by-sample. MOS 4.21. Prohibitively slow: 17 minutes to generate 1 second.
- **WaveGlow (Nvidia, 2019)**: Normalizing flow model. Real-time, 500kHz on GPU. First practical neural vocoder for production deployment.
- **HiFi-GAN (Kong et al., 2020)**: GAN-based vocoder with multi-period and multi-scale discriminators. $167\times$ real-time, MOS 4.40. Became the default vocoder for most TTS systems.
- **BigVGAN (Nvidia, 2022)**: Scaled HiFi-GAN with anti-aliased activation and periodic non-linearity. Universal vocoder: trained once, works on any speaker/domain. MOS 4.45.
- **Vocos (2023)**: Lightweight Fourier-based vocoder. $20\times$ smaller than HiFi-GAN, comparable quality. Preferred for edge deployment.

BigVGAN-v2 is the current gold standard neural vocoder: universal, high-quality, and production-ready. Vocos preferred when model size or latency is constrained.





VALL-E: Neural Codec Language Model for TTS

VALL-E (Wang et al., 2023 – Microsoft)

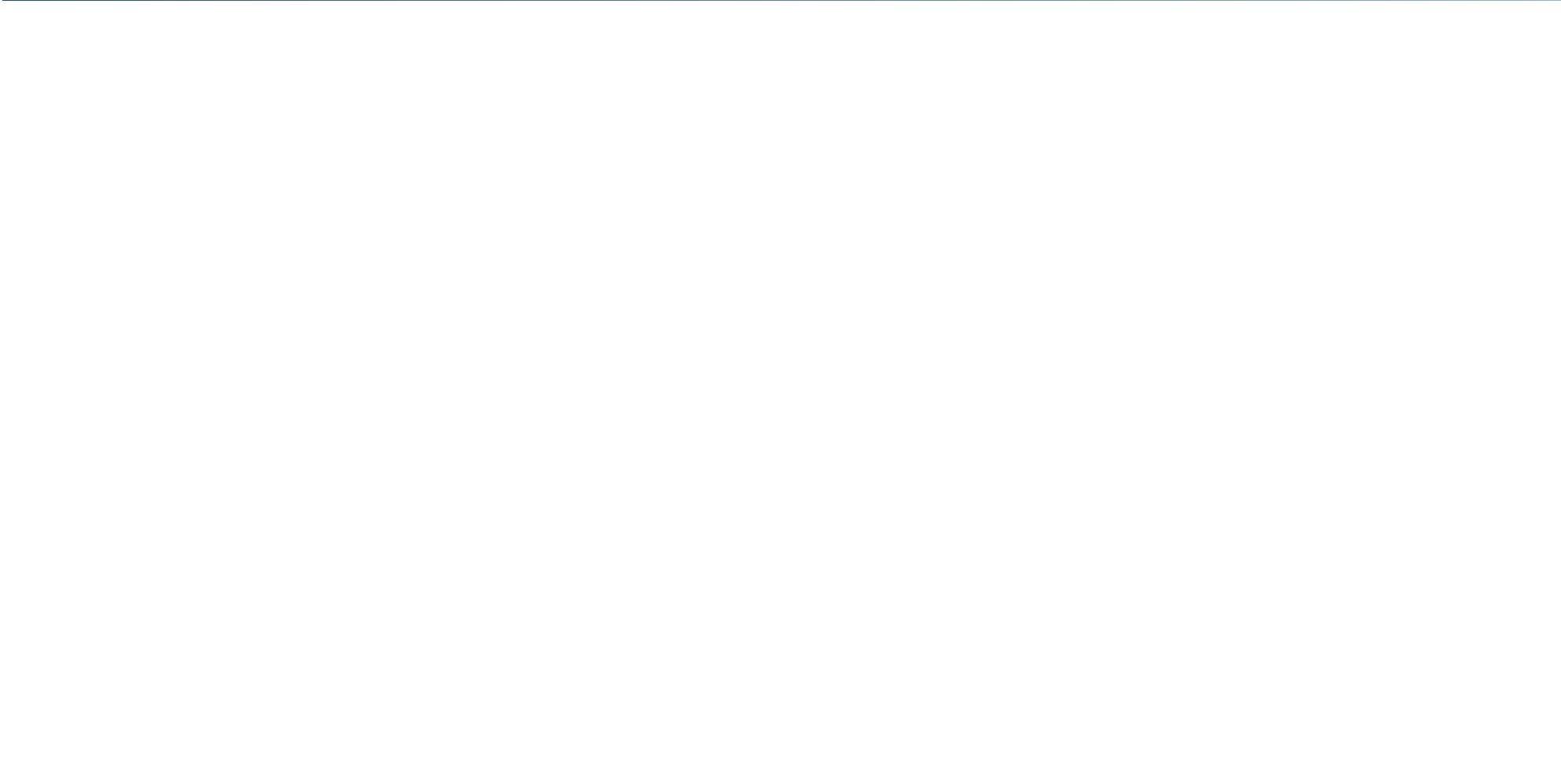
- **Key insight:** TTS is a language modeling problem – predict the next audio token given text and speaker context, just as LLMs predict text tokens
- **Architecture:** EnCodec neural codec tokenizes audio into discrete tokens at 75Hz. GPT-like autoregressive Transformer predicts codec tokens conditioned on phoneme sequence and 3-second speaker prompt.
- **Hierarchical codebooks:** EnCodec uses 8 residual quantization codebooks. VALL-E predicts first codebook autoregressively (coarse prosody), then remaining 7 codebooks in parallel (fine acoustic details).
- **Zero-shot capability:** 3-second speaker prompt achieves voice cloning indistinguishable from the target speaker in a double-blind listening test at 47% accuracy (chance = 50%)
- **Emergent capabilities:** Preserves speaker's emotion, room acoustics, recording conditions from the 3-second prompt – without explicit conditioning



State-of-the-Art TTS Models (2024--2025)

Model	MOS	Cloning	Open?	Architecture
ElevenLabs v2	4.6	Zero-shot	No (API)	Diffusion+LM
OpenAI TTS-1 HD	4.5	No	No (API)	Undisclosed
XTTS-v2 (Coqui)	4.3	Zero-shot	Yes	VITS-based
Voicebox (Meta)	4.4	Zero-shot	No	Flow Matching
NaturalSpeech 3	4.5	Zero-shot	Partial	Diffusion
CosyVoice 2	4.4	Zero-shot	Yes	LM + Flow

CosyVoice 2 (Alibaba, 2024) is the strongest fully open-source zero-shot TTS model, supporting 5+ languages with a 3-second speaker reference and <150ms first-token latency for streaming.



End-to-end models (VITS, XTTS) merge acoustic model and vocoder into a single network. The text frontend (G2P + normalization) remains a separate, rule-based component even in fully neural systems.





GenAI Application: Voice AI Products

Building AI Voice Agents with TTS

- **AI phone agents:** ASR → LLM → TTS pipeline powers automated customer service (Retell AI, Bland.ai, Vapi). Latency budget: <1.5 seconds total for natural conversation.
- **Streaming TTS:** Generate audio in chunks as text is produced by the LLM. First audio chunk in <300ms. Requires sentence-boundary detection to avoid unnatural pauses.
- **Technology stack:** ElevenLabs or OpenAI TTS API for production quality; XTTS-v2 for self-hosted; Vocos vocoder for edge/on-device.
- **Cost optimization:** TTS is typically billed per character. Cache frequently used phrases (greetings, error messages) as pre-generated audio. Can reduce TTS API cost by 40–60%.
- **Latency optimization:** Sentence-level streaming from LLM → parallel TTS synthesis of multiple sentences → audio buffering for smooth playback.

The ASR-LLM-TTS pipeline is the foundation of the voice AI industry – a market projected to reach \$47B by 2030.



GenAI Application: AI Audiobook and Podcast Production

Neural TTS for Content Creation at Scale

- **Audiobook generation:** ElevenLabs and Speechify use neural TTS to narrate books with author's own voice (with consent). Audible uses TTS for 50,000+ out-of-print titles.
- **Podcast creation:** NotebookLM (Google) generates podcasts from documents using two AI voices in conversation – millions of AI podcasts generated since launch.
- **Voice preservation:** Capture voice before illness or death for accessibility or legacy (ALS patients use voice banking with Acapela/VocaliD). Neural TTS makes this practical with only 30 minutes of recordings.
- **Localization:** Dub video content into 30+ languages while preserving the original speaker's voice (HeyGen, ElevenLabs Dubbing) – replaces expensive voice actor recording sessions.

Engineer's Tip

Always include a "speaker verification" step in voice cloning pipelines to confirm the synthesized voice matches the reference embedding before deployment.



GenAI Application: Responsible TTS Engineering

AI Engineers Must Address Voice Deepfake Risks

- **Synthetic speech detection:** Build a discriminator (ASVspoof dataset, RawBoost, RawNet3) to detect AI-generated audio. Deploy as a pre-moderation filter on audio content platforms.
- **Audio watermarking:** SynthID Audio (Google DeepMind, 2023) – imperceptible watermark survives re-encoding, speech codec compression, and noise. Enables provenance tracking of AI-generated audio.
- **Consent verification:** Require speaker consent and signed consent artifacts before voice cloning. Store consent metadata alongside generated audio.
- **Real-time deepfake detection:** Pindrop and Nuance deploy real-time voice liveness detection in call center authentication to detect voice spoofing attacks.

Critical Engineering Responsibility

AI engineers building TTS products have an ethical obligation to implement misuse prevention. Voice deepfakes are used for fraud, election interference, and non-consensual content – technical safeguards are essential.



Key Takeaways

What You Should Remember

- TTS systems combine an acoustic model (text → mel-spectrogram) and a vocoder (mel → waveform). VITS merges both into a single end-to-end model.
- VALL-E reframes TTS as a language modeling problem over neural audio codec tokens – enabling zero-shot voice cloning from 3-second reference audio
- Production choices: ElevenLabs/OpenAI TTS API for quality; XTTS-v2 or CosyVoice 2 for open-source zero-shot cloning; Vocos for edge deployment
- Streaming TTS (<300ms first chunk) is essential for real-time voice AI applications – sentence-level generation is the key engineering pattern
- Responsible deployment requires audio watermarking (SynthID), consent management, and synthetic speech detection as non-negotiable safeguards

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